Markelle Roesti

Website: markellekelly.com • **Email**: markelleroesti@gmail.com

Work Experience

Research Scientist
08/2025 – 11/2026

eBay • Remote

- Developed a toolkit for AI evaluation at eBay, including specialized datasets, benchmarking paradigms, and a human review protocol, covering recommendation, chat, and image generation use cases
- Provided internal consulting, developed policy, and hosted talks and workshops on responsible AI

Applied Research Intern: Responsible AI
06/2024 – 09/2024

eBay • San Jose, CA

- Performed a systematic investigation of bias in LLM behavior in the context of e-commerce, uncovering high-impact, previously-unidentified gender bias issues in multiple models, including GPT-3.5
- Engineered a dataset of targeted counterfactual pairs to detect and characterize these issues

Data-Centric Machine Learning Intern
07/2023 – 10/2023

Apple • Heidelberg, Germany

- Formulated metrics and techniques for the automatic detection of hallucinations in LLM generations
- Developed an interactive tool for the data-driven exploration of such hallucinations and other rare, undesirable LLM behaviors, designed to address the needs of two product teams at Apple

Software Engineer Intern
03/2018 – 06/2020

Project Jupyter • San Luis Obispo, CA

- Implemented new, now widely-used features for an interactive computing environment (JupyterLab), following software development best practices such as automated testing and code review
- Collaborated with the UI/UX team throughout the design process to ensure usability and accessibility

Education

M.S. & Ph.D. in Computer Science
09/2020 – 06/2025

University of California, Irvine • 4.0 GPA

B.S. in Statistics, Minor in Data Science
09/2016 – 06/2020

California Polytechnic State University, San Luis Obispo • 4.0 GPA

Selected Publications

“Variable-based Calibration in Machine Learning Classifiers”

Markelle Kelly & Padhraic Smyth — *AAAI 2023*

“Capturing Humans’ Mental Models of AI: An Item Response Theory Approach”

Markelle Kelly, Aakriti Kumar, Mark Steyvers, & Padhraic Smyth — *FAccT 2023*

“Perceptions of Linguistic Uncertainty by Language Models and Humans”

Catarina Belem*, **Markelle Kelly***, Sameer Singh, & Padhraic Smyth — *EMNLP 2024*

“Benchmark Data Repositories for Better Benchmarking”

Rachel Longjohn*, **Markelle Kelly*,** Sameer Singh, & Padhraic Smyth *NeurIPS 2024*

“Understanding Gender Bias in AI-Generated Product Descriptions”

Markelle Kelly, Mohammad Tahaei, Padhraic Smyth, & Lauren Wilcox *FAccT 2025*

“Bayesian Consensus Prediction for Correlated Human Experts and Classifiers”

Markelle Kelly, Alex Boyd, Sam Showalter, Mark Steyvers, & Padhraic Smyth *ICML 2025*

Teaching and Academic Activities

Workshop Organizer

“The Future of Machine Learning Data Practices and Repositories” at ICLR (2025)

Curator and Librarian

UCI Machine Learning Repository (2020-2025)

Teaching Assistant

Probabilistic Learning (2025), Project in AI (2022), Calculus III & IV (2017-2018)

Reviewer

NeurIPS Datasets and Benchmarks (2022), Cognitive Research: Principles and Applications (2024), EvalEval Workshop at ACL (2026), CHI (2026), NeurIPS (2026)

Fellowships and Awards

HPI Research Center Fellowship *07/2022*

Irvine Initiative in AI, Law, and Society Fellowship *01/2022*

Steckler Center for Responsible, Ethical, and Accessible Technology Fellowship *06/2021*

UCI Computer Science Dean’s Award *09/2020*

Cal Poly Class of 2020 Academic Excellence Award (1 of 2 awarded) *06/2020*

Best Visualization at UCLA DataFest *04/2019*

* denotes equal contribution